

Ryan Zheng

925-201-9539 | ryanzheng@nmokey.com | linkedin.com/in/nmokey | github.com/nmokey

EDUCATION

University of California, Los Angeles

Los Angeles, CA

B.S. in Physics, Data Science Engineering Minor; GPA: 3.875

Sep. 2024 – June 2028

Relevant Coursework: Probability & Statistics, Stochastic Processes, Linear Algebra, Multivariable Calculus, Differential Equations, Discrete Math, Machine Learning, Data Mining

AWARDS & PUBLICATIONS

USA Physics Olympiad (USAPhO) Semifinalist (2023)

top 8% of 5,173 competitors

IMC Prosperity 4 (2026)

top 8% of 18,803 teams

ACM ICPC Break the Binary Medalist (2025)

3rd place of 25 teams

Sigma Pi Sigma Physics Honor Society · ACM AI Projects Director · Jane Street Puzzle Solver

Chen, Y., Jiao, J., & Zheng, R. (2024). Trip generation and built environment impacts: regular vs. essential trips in the U.S. *Proceedings of CICTP 2024*, 3317–3326.

EXPERIENCE

Software Engineer Intern

June 2026 – present

Amazon

Seattle, WA

- Architect a distributed ML experiment-orchestration platform (AWS Step Functions) that compiles declarative configs into fault-tolerant DAGs of multi-node GPU workloads, running unattended multi-day pipelines.
- Design order-independent checkpoint-selection algorithm and parallelize evaluation fan-out to overlap with in-flight training; cut end-to-end runtime by **>50%** and inter-stage handoff latency from **hours to seconds**.
- Build config compiler that deterministically deduplicates state machines, cut experiment setup from **~30 min to <1 min**, streamline research iteration with agentic failure recovery.

Machine Learning Researcher

January 2026 – present

UCLA Scalable Analytics Institute

Los Angeles, CA

- Restructure **3.9 TB** PET/CT archive into a longitudinal panel of **80 subjects** × **4 timepoints**, and evaluate 4 pretrained encoders. Forecast subject-level embeddings one timestep ahead under leave-one-subject-out CV, beating a random-walk benchmark on 66% of held-out transitions while avoiding look-ahead bias.
- Build vision-language model, injecting scan and forecast embeddings as image tokens into a LoRA-finetuned 8B language model with joint regression and classification heads; predict disease outcome **8 weeks ahead** at **0.76 AUC / 72%** vs. **0.58 / 50%** scan-only under matched-budget ablation (10 LOSO runs). Submitting to ICLR.

Machine Learning Intern

January 2025 – January 2026

Scale AI

San Francisco, CA (Remote)

- Built statistical evaluation rubrics and Dockerized agentic RL evaluation environments, supporting industry-standard AI benchmarks (e.g., Aider LLM Leaderboards) across **200+ datapoints** and **6 clients**.

QUANTITATIVE PROJECTS

Perpetual Futures Funding Carry Strategy | *Personal Project*

May 2026 – present

- Backtest funding-rate carry strategy on BTC and SOL perpetuals, find edge survives up to **0.04%** of ADV (\$210K notional) before modeled slippage exceeds funding income; **0.8 Sharpe** net of costs at optimal size vs 2.1 gross.
- Build C++ order book reconstruction and slippage simulation engine processing **400M+** historical events over a **12-month** walk-forward period, decoupling market-data ingestion from simulation via a lock-free queue.
- Calibrate square-root impact model via regression; gate entry on rolling Bayesian AR(1) funding persistence.

S&P500 Return Forecasting | *ACM AI, Hull Tactical Kaggle Competition*

October 2025 – December 2025

- Ensembled daily-horizon forecasting models for S&P500 forward returns using gradient-boosted trees, feed-forward networks, and LSTMs. Achieved **1.48 Sharpe** over 6-month scoring period; **top 22%** of submissions.
- Scaled training across GPUs via gradient checkpointing and 8-bit optimizers, cutting training time by **60%**.

SKILLS & INTERESTS

Technical: C++, CMake, Python, NumPy, statsmodels, pandas, SciPy, PyTorch, R, SQL, Linux

Languages: English (native), Mandarin (fluent)

Interests: Skiing, tennis, jazz piano, competitive archery, game speedrunning (record holder), poker